



Student Use of Keywords and Limiters in Web-scale Discovery Searching



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ABSTRACT

Using transaction log analysis of student search histories in EBSCO Discovery Service, which we branded RVOneSearch, we seek to answer: 1. Do students use the limiters provided in RVOneSearch? 2. How effectively do students use keywords in RVOneSearch after receiving instruction on keywording? In Spring 2012, RVOneSearch became an integral part of our 80-minute librarian-led sessions that are required of all English Composition I courses. We began focusing our instruction on selecting appropriate keywords when we noticed that regardless of interface, students struggled most with identifying the right search terms. With RVOneSearch we stopped teaching multiple interfaces and instead began teaching one reference database for background and context, and RVOneSearch for scholarly sources. We spend less time on the nuances of interfaces and more time on evaluating results. We also teach students to use the facets and limiters available in RVOneSearch. In this study we wanted to determine if students used the available facets and limiters and whether they chose appropriate keywords after instruction on how to do so. Our results inform how and what we teach students and are applicable to others teaching with web-scale discovery.

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INTRODUCTION

Web-scale discovery services are becoming an integral component of academic libraries. Seeking to provide students with an enticing alternative to Google for their research needs, librarians have been implementing “one-stop searching” services since the early days of federated search. However, these tools proved to be cumbersome and slow with poor relevancy ranking. With the introduction of Google Scholar in 2004, Breeding (2005) declared that libraries’ “strategy of metasearch that depends on live connections casting queries to multiple remote information sources” could not compete with the speed, simplicity, and relevancy ranking of systems based on centralized indices (p. 27). Vendors responded and began offering full-service discovery products like Summon (Serials Solutions), EBSCO Discovery Service, Primo Central (Ex Libris) and WorldCat Local (OCLC). These tools meet the need—be it real or perceived—for single search boxes on library websites that give students one-stop access to many of their library’s resources.

The Evelyn S. Field Library at Raritan Valley Community College implemented EBSCO Discovery Service (EDS) in September, 2012, and rebranded the tool as RVOneSearch. Instruction librarians began teaching RVOneSearch in the spring, 2013 semester as an integral part of our Information Literacy I sessions, which are 80-minute librarian-led sessions required of all English Composition I (ENGL 111) courses. We recognize that there are a number of pedagogical limitations to one-shot

information literacy instruction. These limitations are compounded by students’ desire to find sources quickly and easily, their over-reliance on Google searches to find all types of information, and the satisfying search behavior that is characteristic of information seekers (Simon, 1979), especially college students (see for example Barrett, 2005; Connaway, Dickey, & Radford, 2011; Prahba, Connaway, Olszowski, & Jenkins, 2007; Warwick, Rimmer, Blandford, Gow, & Buchanan, 2009). Historically, we have taught students in ENGL 111 how to search the catalog and two or three other databases. The majority of class time was spent demonstrating search techniques in the various interfaces. Over time, the librarians moved away from a focus on tool instruction and began spending more time on selecting appropriate keywords. We noticed that regardless of search interface, students struggled most with identifying the right terms to find the sources they needed. Implementing RVOneSearch helped our efforts because we decided to stop teaching multiple interfaces to ENGL 111 students and instead teach one tool for background information (CQ Researcher, Credo Reference, or a Facts on File database, for example) and RVOneSearch for all other sources. Now, a significant portion of class time is spent working with students on refining their topics and identifying keywords. Many students have some familiarity with the EBSCO search interface because EBSCO products are used in a number of the area high schools. When teaching RVOneSearch, we spend less time on the nuances of the interface and more time on selecting keywords and evaluating search results. Because narrowing results is an important skill when using a discovery system, the librarians also teach students how to use the facets and limiters that are available in RVOneSearch.

We determined that we could evaluate student search behavior in RVOneSearch by analyzing Search Histories, which are stored by

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RVOneSearch during a user's search session. We wanted to answer the questions:

1. Do students use the limiters provided in RVOneSearch?
2. How effectively do students use keywords in RVOneSearch after receiving instruction on keywording?

Around the same time that we undertook this research, [Badke \(2013\)](#) asserted that discovery systems encourage students to pursue the path of least resistance when it comes to research:

You can't tell me that even 5% of users actually facet their discovery searches... They just muddle through whatever result their initial keyword searches got them. (p. 66).

Our research uses search log analysis to determine if this is true of students taking their first college composition course.

LITERATURE REVIEW

In 2005, Breeding proposed that libraries begin making the switch to metasearch tools that rely on centralized indices rather than on distributed search (p. 29). Within a couple of years, vendors began responding to this type of transition by offering web-scale discovery products. The introduction of this promising new technology to meet students' desire for easy information retrieval and librarians' desire to get library resources into students' hands has produced a flurry of literature related to the selection, implementation, usability, and instruction of web-scale discovery products. In January, 2011, *Library Technology Reports* devoted an entire issue to web-scale discovery, including an explanation of what web-scale discovery is and why it is important, and individual chapters devoted to the big four that had emerged by that time: OCLC WorldCat Local, Serials Solutions Summon, EBSCO Discovery Services, and Ex Libris Primo Central ([Vaughan, 2011](#)). Numerous library trade magazines and journals have offered descriptions and comparisons of specific products ([Birong, Kelley, & Garrison, 2009](#); [Chickering & Yang, 2014](#); [Fernandez, 2011](#); [Rowe, 2011](#); [Vaughan, 2011](#)), explanatory articles to serve as web-scale discovery primers for librarians ([Little, 2012](#); [Notess, 2011](#); [Wisniewski, 2014](#)), and speculations on the potential value or impact of web-scale discovery to libraries ([Richardson, 2013](#); [Vaughan, 2012](#); [Way, 2010](#)).

Several usability studies have been conducted and the findings relate to the questions investigated in this study. Usability tests have revealed that participants are either unaware of or choose not to use limiters and facets ([Dalal, Kimura & Hofmann, 2015](#); [Gross & Sheridan, 2011](#)), cannot understand which limiters would improve their results ([Cassidy, Jones, McMain, Shen, & Vieira, 2014](#)), or are more likely to use advanced search, limiters, and facets for open-ended searches rather than when searching for known items ([Hanrath & Kottman, 2015](#); [Nichols, Billey, Spitzform, Stokes, & Tran, 2014](#)). [Williams and Foster \(2011\)](#) found that all but five of six participants used a post-search limiter or refinement when searching EBSCO Discovery Service although they were used sparingly, in contrast to the participants of a follow up study who used post-search refinements more frequently in Serials Solutions' Summon ([Foster & MacDonald, 2013, p. 8](#)).

[Gross and Sheridan \(2011\)](#) and [Dalal, Kimura, and Hofmann \(2015\)](#) discovered that most participants failed to use appropriate keywords to articulate their information needs. [Johnson \(2013\)](#) shares that even a faculty member could not appropriately keyword an information need in a single-search box; the faculty member commented he "would just try Google" as his next step, rather than alter search terms or try one of the library's subject-specific databases (p. 73). [Johnson](#) concludes that although users like the single-search box interface, they are ineffective at refining a search for better results. [Asher, Duke, and Wilson \(2013\)](#) compared student search behaviors in EDS, Summon, Google Scholar and conventional library resources and found that students

searching EDS used the highest quality resources to complete given tasks. Regardless of the tool, though, "students treated almost every search box like a Google search box, using simple keyword searches in 81.5 percent of the searches observed" (p. 473). Thus large numbers of results are obtained, students have difficulty evaluating the quality of results, and they put a lot of trust in the search tools' relevancy-ranking and markers of authority (peer reviewed or "scholarly" designations).

Another area related to the current study is the impact of web-scale discovery tools on information literacy instruction (ILI), but fewer research studies have been conducted on this topic. Authors have speculated on how web-scale discovery tools support or will change ILI ([Cmor & Xin, 2012](#); [Fagan, 2011](#); [Fawley & Krysak, 2012](#)) or have related their own personal experience teaching a web-scale discovery tool ([Azadbakht & Polacek, 2015](#)). In a conceptual article, [Seeber \(2015\)](#) argues that particularly in an era of web-scale discovery, librarians should be teaching the process of information creation, not how to search an interface. [Buck and Steffy \(2013\)](#) use the results of interviews and surveys of teaching librarians to propose promising practices for teaching web-scale discovery tools.

According to [Dalal, Kimura, and Hofmann \(2015\)](#), findings from their usability study have informed changes to their library's information literacy instruction. They found that students lack an understanding of the scholarly publishing process and the purpose of a discovery tool, and that students fail to use advanced search techniques or locate the full text of articles. Knowing these challenges, they conclude that they need to "teach more slowly and with more repetition to ensure real mastery of even the most basic concepts" (p. 675). Our research reveals similar findings about the impact of web-scale discovery on ILI.

METHODS

Data were collected in the spring semester of 2013 from 118 students enrolled in 27 sections of ENGL 111. After conducting their own independent research using RVOneSearch, students were verbally instructed by the teaching librarian to print out their Search History. Students were informed that printing out their Search Histories was optional and that they were not obligated to share them with us, and student names were not identified on the Search Histories. Since Raritan Valley Community College has no Institutional Review Board (IRB), approval could not be obtained. However, no identifying data were collected that could be traced back to any individual student.

By default, a Search History is not visible in RVOneSearch but a user can opt to make it visible by clicking the Search History link directly below the search box. When visible, individual searches in a history can be selected and combined using Boolean connectors. The results of a prior search can be viewed, and prior searches can be revised from the Search History. A Search History can also be printed ([Fig. 1](#)).

The number of searches varied from section to section of ENGL 111 and across individual students within sections. The number of searches students performed ranged from 1 to 26 with the average number of searches as 5.9.

Student Search Histories were analyzed based on two broad criteria:

- Use of EBSCO provided limiters
- Student keyword abilities.

EBSCO Discovery Service provides a number of limiters that can be applied before or after conducting a search. Our librarians teaching IL instruction for ENGL 111 normally only teach one or two of these limiters so as not to overwhelm the students. The most commonly taught limiter is 'Peer Reviewed' because nearly all ENGL 111 instructors require students to use at least one peer reviewed article as a source. This and other limiters appear on the left side of the results screen. When a limiter is selected, the search is modified and new results appear that

Search History/Alerts

[Print Search History](#) | [Retrieve Searches](#) | [Retrieve Alerts](#) | [Save Searches / Alerts](#)

☐ Select / deselect all		Search with AND	Search with OR	Delete Searches	Refresh Search Results
Search ID#	Search Terms	Search Options		Actions	
☐ S2	 obesity and fast food	Limiters - Peer Reviewed Expanders - Also search within the full text of the articles Search modes - Find all my search terms		View Results (32,791) View Details Revise Search	
☐ S1	 obesity	Limiters - Peer Reviewed Expanders - Also search within the full text of the articles Search modes - Find all my search terms		View Results (534,012) View Details Revise Search	

Fig. 1. Search History view.

reflect the additional search condition. The modified search appears on the Search History as an additional search.

We examined Search Histories for the use of the following EBSCO provided limiters:

- Peer reviewed
- Date
- Language or geography
- Advanced search Boolean¹
- Subject term (narrow by two-letter search tag: DE, SU or ZU²).

Coding for the EBSCO provided limiters consisted of marking a 1 for yes, the student used this limiter, and 0 for no, the student did not use this limiter.

We also examined student keyword abilities to determine how effectively students applied what they were taught about keywords during the librarian-led information literacy session. We examined student Search Histories for five common keywording problems:

- Misuse of quotation marks: coded 1 if there were no misused quotation marks, 2 if quotes were misused once, and 3 if quotes were misused more than once
- Misuse of phrases: coded 1 if there were no misused phrases, 2 if there was one misused phrase, and 3 if there was more than one misused phrase
- Repetitive spelling errors: coded 1 if there were no spelling errors, 2 if there were spelling errors that were noticed by the student and corrected, and 3 if there were repeated spelling errors that were not corrected
- Inappropriate keywords: coded on a scale 1 through 5, with 1 indicating all keywords were appropriate and 5 indicating that the majority of keywords were inappropriate
- Lack of variance in keywords: coded on a scale 1 through 5, with 1 indicating the student used good variance in keywords and 5 indicating the student did not vary his/her keywords at all.

The final category used to evaluate each search was “Demonstrated search improvement” (or lack of improvement) from the beginning of the search to the end. “Demonstrated search improvement” was coded on a scale from 1 through 4: a 1 was assigned if the student had

an overall effective set of search terms from the start; 2 was assigned if the student's search improved significantly throughout the Search History; 3 was assigned if the student's search showed slight improvement; and 4 was used if there was no improvement in the overall quality of the search terms throughout the Search History. For this criteria, we were looking for any indication that students used keyword techniques taught in the class to modify their searches.

LIMITATIONS

This research has some limitations. Because we only evaluated search logs, we do not know whether or not students found the sources they needed for the assignment, regardless of the quality of their searches. It is possible that the searches we identified as poor actually got a student to the articles he or she needed. Also, because of the dynamic nature of the classes we teach, the amount of time students spent during their independent research portion varied from class to class. Potentially, students could have applied more refinements to their searches if they had more time. Finally, because we collected the data during actual instruction sessions, the librarians and faculty members were assisting students throughout the search process. Further qualitative research is needed to supplement this quantitative data and determine if students' search strategies, whether effective or ineffective by librarian standards, are leading them to the resources they need.

RESULTS

USE OF EBSCO PROVIDED LIMITERS

With the exception of the Peer Reviewed limiter and the Date limiter, a very small percentage of students used the EBSCO provided limiters in RVOneSearch. Table 1 indicates the number of students who used the limiter (Yes) or did not use the limiter (No), and the percentage of students who used the limiter (% Yes, n = 116).

Fewer than half (41%) of the students used the Peer Reviewed limiter, although all students were required to find a peer reviewed article for their research paper. Using the Peer Reviewed limiter is a

Table 1
Use of limiters and facets in RVOneSearch.

	Subject-term (narrow by, DE, SU or ZU)	Advanced search Boolean	Limit language or geography	Limit date	Limit peer reviewed
Yes	14	17	4	33	47
No	102	99	112	83	69
% Yes	12%	15%	3%	28%	41%

¹ We determined that a student modified his/her search in the Advanced Search screen if the Search History showed terms connected using AND, OR, or NOT in all caps. Boolean connectors in all caps are auto-populated when search terms are entered in multiple search boxes on the Advanced Search screen.

² EBSCO Discovery Service populates the following into the search box based on user selection and which database index is providing results. These are known as “two letter search tags.” DE = Heading OR keyword SU = Subject (word indexed) ZU = Headings.

strategy the authors emphasize in their instruction. The Date limiter is not taught as consistently but is mentioned at some point in most information literacy sessions. Only 28% of students used this limiter.

The other limiters analyzed were minimally used, but also were not explicitly taught unless we recognized the need to use the limiter while helping an individual student. Narrowing by subject-term (using the hyperlinked subject-terms on individual articles) was only done by 12% of students. This strategy is explicitly taught in English Composition II (ENGL 112) classes, but not in the ENGL 111 classes studied here. The Advanced Boolean Search screen is not set to default in the RVOneSearch settings so students would need to click on the 'Advanced Search' link in order to access this feature. The Advanced Search link was used by 15% of students to add terms to their search. Only 3% of students used either the language limiter or the geography limiter. Interestingly, one student limited his search results to the Albanian language.

STUDENT KEYWORD ABILITIES

This study analyzed a variety of keyword abilities. Because the scoring rubric for each ability is different, the results for each keyword-specific ability are discussed separately.

MISUSE OF QUOTATION MARKS

Searches were given a score of 3 if quotation marks were misused more than once; 2 if quotation marks were misused once; and 1 if there were no errors in the use of quotation marks. After coding the data, we realized a score of 1 included both searches that used quotation marks correctly and searches that did not use quotation marks at all. We did a separate count for searches that used quotes correctly and deducted the number from the total number of searches that were scored with a 1. Only six students (6%) used quotations incorrectly more than once, and the same number used quotations incorrectly one time. Eight students (7%) used quotation marks correctly. Most students (89%) did not use quotation marks at all, even if doing so would have improved the search results. This search strategy is taught on an as-needed basis in most ENGL 111 classes. However, the number of students who used quotation marks correctly (8) is fairly close to the number who used quotation marks incorrectly (12 total). For those who did use quotation marks, we were interested in whether or not students understood the function of quotation marks when searching. A number of students are already familiar with the strategy from previous search experiences either from using high school databases or general web searching. Examples of students' misuse of quotation marks include:

mobile phone "texting" (Search 4)
 "is college for everyone" (Search 44)
 obesity, "beauty" in women in other cultures (Search 46)
 obesity and "government" (Search 86).

The inclination to put individual search terms in quotations demonstrates a serious lack of understanding of the purpose of quotes when searching.

MISUSE OF PHRASES

Searches were given a score of 3 if more than one inappropriate search phrase was used; 2 if one inappropriate search phrase was used; and 1 if only appropriate phrases were used. Because the proper use of quotations was considered separately, we did not analyze whether or not the phrases were placed in quotation marks when determining appropriateness. One inappropriate phrase was used in 28% of searches, and 21% of searches used more than one inappropriate phrase. However, slightly more than half the searches (52%) either did not use phrases or used phrases that were considered appropriate for searching. Given this number, it seems that more students probably should have been

using quotation marks to identify phrases in their searches. Typically, the misuse of phrases occurred when students entered full statements or questions into their searches rather than keywords. For example:

body image and *different cultural women* (Search 82)
 technology and *advances in the past 10 years* (Search 24)
 internet impairing ability to think and learn (Search 29)
 biased against the elderly driving (Search 109).

Such searches suggest to us that these students do not grasp the keyword concepts that we spend a lot of time working on at the beginning of the session.

REPETITIVE SPELLING ERRORS

Spelling errors can cause real problems for students searching a web-scale discovery tool. Students are accustomed to spell-check and the "Did you mean..." feature of web search tools to correct their spelling mistakes. RVOneSearch does not provide students with suggested spelling corrections. If a student is not paying attention to a misspelled keyword, or is not aware that he/she has misspelled a term, their results will be limited and potentially irrelevant. Searches were given a score of 1 if there were no spelling errors; 2 if there was a spelling error that was corrected; and 3 if a spelling error was not corrected. Only a small percentage of searches (5%) contained errors that were not corrected and 14 searches (12%) had a spelling error that was immediately corrected in the next modified search. The majority of searches (83%) contained no spelling errors. Examples of spelling errors that did occur include:

comunity colleges and outcomes; *comunity* colleges and employment (Search 7)
 "America dream" (Search 45, repeated in 7 search modifications)
 beauty and *pre* pressure (Search 25, repeated in 3 search modifications).
*cooperate*³ racism; *cooperate* glass ceiling; social inequality in *cooperate* world (Search 67).

Again, the biggest concern with spelling issues is when a student does not recognize the error—as in Search 67—and continues down a path of results that are not relevant or assumes there is no information on his/her topic.

INAPPROPRIATE KEYWORDS

Overall searches were scored on a scale of 1 to 5 based on the appropriateness of the keywords used in the search. Searches that used all appropriate keywords were given a score of 1 and searches in which the majority of keywords were inappropriate were given a score of 5. The mean score was 1.96; mode was 1. These scores indicate that the majority of searches used appropriate keywords throughout the search. This was one of the more indicative criteria as to whether or not students understood appropriate searching in RVOneSearch, particularly based on the librarian-led instruction. For this reason, overly broad search terms, the essay titles upon which the student's research was based, and colloquial terms were considered inappropriate, among others. Examples of inappropriate keywords we identified include:

television and *dumb* and reality (Search 20)
 video games and *dumbing down* (Search 20)
 obesity *blame yourself* (Search 79)
 the US penal system (Search 99)
picky eaters AND children (Search 37).

Where students' keywords were inappropriate, they often failed to consider synonyms or more formal terms that would be likely to appear

³ Should be *corporate*.

in college-level sources, and instead searched colloquial or informal terms that are part of their everyday vernacular.

VARIANCE IN KEYWORDS

Searches were analyzed to determine if the student varied his/her keywords throughout the search process. In other words, we were looking for evidence that the student applied our instruction on keywording, which emphasizes brainstorming synonyms, broader terms, narrower terms, and related terms. Primarily, we were looking for whether students added additional keywords to overly broad searches and whether they changed some keywords throughout the search session. For example, if a student began a search using “teenagers” as a keyword, we were looking to see if “adolescents” or “young adults” was later used in the search. Searches were scored on a scale of 1 to 5, with a score of 1 indicating good variance in keywords and a score of 5 indicating no variance of keywords. Scores of 5 were also given to searches that “keyword hopped”: the student simply tried several different keywords that were completely unrelated.⁴ The mean score was 3.5; mode was 3. The majority of searches scored in the middle of the ranking. Search 9 provides an example of a student who repeatedly used the same keywords without considering synonyms or related terms that may have retrieved better results (below represents the student’s entire Search History):

fast food overweight health problems
 fast food overweight health issues (*changed “problems” to “issues”*)
 fast food obesity health issues NOT kids (*added Boolean “NOT kids”*)
 fast food obesity health problems NOT adults (*reverted to “problems” and changed “kids” to “adults” — it’s unclear which results the student wanted to eliminate by using NOT*)
 fast food obesity health problems NOT kids AND kids (*not sure if student understood how Boolean would group these terms*).

Search 14 shows a student’s attempt to modify a search without ever considering if the term “american [sic] dream” was not the appropriate search term to use:

The american dream
 The american dream dead or alive
 problems or issues with the american dream (*over 16 million results; “or” functioned as Boolean operator without student’s knowledge*)
 TI problems or issues with the american dream (*searched the phrase as a Title*)
 The american dream (*reverted back to the original search phrase, limited to Peer Reviewed, and ended with 231,754 results*).

DEMONSTRATED SEARCH IMPROVEMENT

Overall searches were scored on a scale of 1 to 4 based on whether or not general improvement was observed during the search progression. Searches that were keyworded properly and effectively from the start were given a score of 1. Searches that demonstrated significant improvement were given a score of 2. Searches that demonstrated only slight improvement were given a score of 3. Searches that showed no improvement were given a score of 4. The mean score was 3.22; mode was 3.5. The majority of searches were given a score of 3 or 4, showing only slight or no improvement at all.

For example, in Search 111, the student tried variations on the same keywords, seemingly thinking that word order in the search would affect results:

concussions in children playing football
 children and concussions with football

children and football related concussions
 youth football concussions.

The student in Search 3 demonstrates a similar problem. He/she repeatedly uses the phrase “childhood obesity,” pairing it with different keywords and often reverting back to the previous search.

childhood obesity lack of exercise
 childhood obesity lack of exercise bmi public health
 childhood obesity lack of exercise (*reverts to original search*)
 JN “Academic Pediatrics” (*clicked a Journal title hyperlink which ran a new Journal Title search*)
 childhood obesity lack of exercise (*back to original search phrase*)
 “reduced genetic influence”
 childhood obesity overeating
 JN “Clinical Therapeutics”
 childhood obesity overeating (*student never thinks to combine his search phrase with the Journal title search*)
 childhood obesity over indulging
 DE “Parents – Attitudes” (*clicked a subject heading hyperlink which ran a new search*)
 childhood obesity “over indulging” (*reverts to previous search phrase but adds ineffective quotes*)
 “childhood obesity over indulging”
 childhood obesity over indulging (*remove quotes, back to previous search phrase*).

Although this student comes close to improving his search, he/she never quite gets there. For example, if he/she had included the subject term “Parents—Attitudes” with “childhood obesity,” perhaps he/she would have found more specific results.

DISCUSSION

When looking at the results for specific keywording abilities, it seems that many of the students applied concepts they were taught in class. However, analysis of overall searches indicates that despite using specific techniques, students are not using their keywords as part of an effective search strategy. For example, in Search 28, the student began with “globalization and secret societies,” then removed the term “globalization” from the search, next searched for “bilderberg group” and finally ended with “resistance to globalisation” (note the changed spelling). The individual keywords used received good scores for appropriateness and variety, but the search received a low score for improvement because the keywords were never combined in a way that could have refined the results effectively.

Other searches demonstrated this lack of strategy, where students just keyword-hopped from one term or phrase to the next without trying to combine search terms for better results. In Search 16, the student started with “Sammy Sosa,” then tried “barry bonds,” followed by “HGH,” then “HGH baseball,” “cheating baseball,” “1990 baseball,” and finally “home runs baseball.” Presumably the student came across some kind of information that indicated 1990 was a turning point in the issue, but the search itself is still quite unfocused. The keywords themselves are not the worst (although not the best either), but each modified search varied so widely from the previous that it seems the student had no idea what he/she was really looking for. This behavior supports Dalal, Kimura, and Hofmann’s (2015) findings that when keyword searching, students would simply change their search terms when they were unsuccessful and that “most often, their new choice for search terms seemed little more than a guess” (p. 671).

Both of the above examples could be considered in light of Project Information Literacy’s findings around students’ need for research contexts. Head and Eisenberg (2009) outline a typology of contexts to explain the consistent struggles students have with academic research. One context they found students struggling with is language: “Language

⁴ For example, in Search 16, each search used a different keyword or phrase: Sammy Sosa; barry bonds [sic]; HGH; HGH baseball; cheating baseball; 1990 baseball; home runs baseball.

context involves becoming more comfortable with the language, terms, and discourse of a topic area... Without a context for the vocabulary and terminology of a topic—what things are called and what they mean—students could not proceed with any confidence and reported that they usually did so with little success” (Head & Eisenberg, 2009, p. 8). Our students’ keyword hopping and experimentation with keywords that seem to offer little chance for success suggest that they lack the appropriate language context for their research needs. They are unable to strategically articulate search terms that will retrieve the information they need. Instead, they more or less play with a variety of keywords hoping to land on a few that generate satisfactory results.

This lack of language context results in poor understanding of what constitute appropriate or effective keywords. For example, the student in Search 7 began with a very general search for “community [sic] colleges and college degree,” changed keywords in the next search to “2 years education and four years education,” tried “community [sic] colleges and employment” in the fifth search modification, and ended up with “community [sic] colleges and outcome education” as the final search phrase. This student varied keywords throughout the seven search modifications, but persisted in spelling “community” incorrectly, couldn’t seem to decide between using numerals (“2”) or spelling out numbers (“four”) so tried both, and never really pinned down the keyword to describe the information needed (was it education, employment, or outcomes?).

We selected the particular keyword strategies analyzed in this study based on our informal observations of problems students encounter when we help them one-on-one. Our findings indicate that some of these problems are not as widespread as we initially thought. Despite the fact that we usually teach the use of quotation marks at some point in most sessions, only 11% of students in this study used quotations at all. Of those that did use quotes, eight students used them correctly and only 12 did so incorrectly. However, our finding that 52% of students used appropriate search phrases suggests that more students could probably be using quotation marks to refine their searches. Still, 28% used one inappropriate search phrase and 21% used more than one inappropriate search phrase. The more pressing problem is students’ inability to identify proper phrases to use as search terms, not whether they think to put those phrases in quotes.

We are often concerned that auto-correct spelling features have affected students’ ability to recognize their own spelling errors in library search tools. In fact, EBSCO recently followed Google’s lead by implementing an auto fill feature that would populate suggested search terms and phrases as a user typed. Our findings indicate that repetitive spelling errors such that would drastically affect a student’s results were present in only 5% of the searches. Although the auto fill feature was not available at the time of the study, when it did become available, the authors disabled it in RVOneSearch because the suggested searches are often no better (or are worse) than the searches students develop on their own and the suggested searches still contain spelling errors created from previous users’ searches. The auto fill feature does not address students’ spelling or other keywording problems.

Returning to Badke’s (2013) guess that fewer than 5% of users facet their searches, our findings reveal that more than that number do use limiters and facets. While the numbers are not drastic, 41% of our students used the Peer Reviewed limiter, 28% limited by date, and 12% made use of available subject terms to narrow their search. Although lower than the 29% found by Asher et al. (2013), we were pleased to see that 15% of our students ended up performing advanced searches, even though RVOneSearch defaults to a single search box and we do not regularly teach students to select the Advanced Search option. Not surprisingly, only 3% of students used a language or geography limiter, but there were four students who noticed that these options existed among the search limiters.

CONCLUSION

This research was undertaken to investigate two questions:

1. Do students use the limiters provided in RVOneSearch?
2. How effectively do students use keywords in RVOneSearch after receiving instruction on keywording?

We conclude that students will most likely use a limiter or facet that will lead the students to sources that meet assignment requirements. We saw that the highest percentage of students apply the Peer Reviewed limiter; the ENGL 111 research paper requires students to use at least one peer reviewed article, so this limiter was the most relevant to the needs of their assignment and is a limiter that is consistently emphasized by the librarians.

Despite the fact that we spend a significant portion of class time on selecting appropriate keywords, our research reveals that students generally either select poor keywords to articulate their information needs, or when they do select appropriate keywords, lack an understanding of how best to combine search terms to refine results. We would expect similar results regardless of the search tool used. Because our students are using a web-scale discovery tool, the most ineffective searches result in an unmanageable number of results. In searches where students effectively combined multiple search terms and applied at least one limiter, the results were much more manageable, suggesting that instruction on these methods can help students obtain better results in a web-scale discovery tool. Future research may include qualitative analysis of whether or not students are finding the sources they need despite their keyword selection and the often unmanageable number of results they retrieve in web-scale discovery. Do their methods work for them, even though librarians would judge their search abilities to be lacking?

Around the time we undertook this research, Buck and Steffy (2013) surveyed and interviewed librarians about promising practices for teaching a web-scale discovery tool. Their list of promising practices is effective instructional strategies for teaching nearly any information literacy concept; there is little about them that is specific to teaching web-scale discovery. Similarly, our research reveals that the same strategies that are necessary when teaching students to search for information in any electronic venue apply to web-scale discovery searching. Students need help selecting appropriate keywords, developing a search strategy that makes the best use of those keywords, and understanding that search platforms provide tools (limiters, facets, subject-term links, advanced search screens, etc.) to help them refine their results. Our findings support the best practices suggested by Fawley and Krysek (2012) to “focus on developing search terms” and “teach limiters or facets” (pp. 211–212). As Asher et al. (2013) state, “well-prepared students can effectively use a variety of search tools, while poorly prepared students will likely struggle even with the best-designed tools” (p. 476). While we still believe that we should not spend a lot of time teaching specific interfaces, our research finds that students are more likely to use refinement tools when they are explicitly introduced to ones that have an immediate impact on helping them find the types of sources they require. Our instruction needs to prepare them to apply these strategies to any search platform.

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